

Problem Statement 1

4.Code written in Python.

Code:

```
bike = input("Enter the bike you want : ")
model = input("Enter the Model : ")
colour = input("Enter the Colour : ")

if bike == "Hardly Davidson" or bike == "hardly davidson" or bike == "HARDLY DAVISON":
    if model == "Roadster":
        if colour == "Red" or colour == "Blue" or colour == "Black" or colour == "Navy Blue":
            print("Cool Bike! It's really good for long travel and mountain ride")
        elif colour == "Orange" or colour == "Violet" or colour == "Green":
            print("Great bike It's really awesome")
        else:
            print("We didn't have the color of that bike")
    elif model == "Rpg1250":
        if colour == "Red" or colour == "Blue" or colour == "Black" or colour == "Navy Blue":
            print("Great really good")
        elif colour == "Orange" or colour == "Violet" or colour == "Green":
            print("It's so cool")
        else:
            print("We didn't have the color of that bike")
    else:
        print("We didn't have that model it will come next week.Sorry for the inconvenience")
elif bike == "Royal Enfield" or bike == "royal enfield" or bike == "ROYAL ENFIELD":
    if model == "Classic 350":
        if colour == "Red" or colour == "Blue" or colour == "Black" or colour == "Navy Blue":
            print("It's one of my favourite bikes in my childhood")
        elif colour == "Orange" or colour == "Violet" or colour == "Green":
            print("I literally like the bike because of its model and its function")
        else:
            print("We didn't have the color of that bike")
```

```

elif model == "Classic 500":

    if colour == "Red" or colour == "Blue" or colour == "Black" or colour == "Navy Blue":

        print("Cool bike")

    elif colour == "Orange" or colour == "Violet" or colour == "Green":

        print("Great")

    else:

        print("We didn't have the color of that bike")

else:

    print("We didn't have that model it will come next week.Sorry for the inconvenience")

else:

    print("Sorry for the inconvenience we didn't have the model and colour that you have requested")

```

Output:

Enter the bike you want : Hardly Davidson

Enter the Model : Roadster

Enter the Colour : Orange

Great bike It's really awesome

5.Code Wriiten in SQL

Code:

```
CREATE DATABASE expense_tracker;
```

```
USE expense_tracker;
```

```
DROP TABLE IF EXISTS expenses;
```

```
DROP TABLE IF EXISTS users;
```

```
CREATE TABLE users (
```

```
    user_id INT AUTO_INCREMENT PRIMARY KEY,
```

```
    username VARCHAR(50) UNIQUE NOT NULL,
```

```
    email VARCHAR(100) UNIQUE NOT NULL,
```

```
    password VARCHAR(255) NOT NULL,
```

```
    budget FLOAT DEFAULT 0
```

);

```
CREATE TABLE expenses (  
    expense_id INT AUTO_INCREMENT PRIMARY KEY,  
    user_id INT NOT NULL,  
    expense_date DATE NOT NULL,  
    category VARCHAR(50) NOT NULL,  
    description VARCHAR(100),  
    amount FLOAT NOT NULL,  
    FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE  
);
```

Problem Statement 2

1.Where would you rate yourself on (LLM, Deep Learning, AI, ML). A, B, C [A = can code independently; B = can code under supervision; C = have little or no understanding]

Ans:B – can code under supervision

I know some of the concepts in ML Workflow but need an guidance to work with the development.

2.What are the key architectural components to create a chatbot based on LLM? Please explain the approach on a high-level

Ans:

I would use RAG Concept

RAG – Retrieval Augmented Generation

RAG is nothing but consist of information based on the document like pdf,etc.

Real Case Scenario is MNC Companies developed Chatbots which are the applications of RAG.

Large Language Model -> consist of Billions of Data

Architectural Components:

1.Query – Loading all the Information

2.Chunking – separating information to small pieces

3.Embeddings-small pieces of information converted to numerical values and vectors

4.Vector Database – consist of information in the format of numerical values in a list.Stored in it like graph(both X and Y axis).Can retrieve information quickly.

- 5.Retrieval – Similarity Search(use Cosine Similarity)
- 6.Vectors – Consist of Information in Numerical Value like 0.78,0.12,etc
- 7.Rearranging -ReOrdering the Retrieve information.
- 8.Prompt + Context – User gives prompt to chatbot.
- 9.LLM – Retrieving information based on the User Request.Eg:ChatGPT by OpenAI,Claude by Anthropic.

3.Please explain vector databases. If you were to select a vector database for a hypothetical problem (you may define the problem) which one will you choose, and why?

Ans:

A Vector Database is a Database used to store values in Numerical Format like 0.34,0.87,etc. Unlike Traditional Database which is use to retrieve value based on the data,Vector Database can retrieve information much more faster and find similarity in the data.

Applications:Generative AI Chatbots:ChatGPT,Claude,Deepseek,Gemini,etc.

Real World Example:Chatbot

Imagine a company has 1,000 PDF documents.

A user asks:

"How many jobs available in the company?"

The Vector Database searches the document and retrieves the information with similiarity and then the LLM generates it